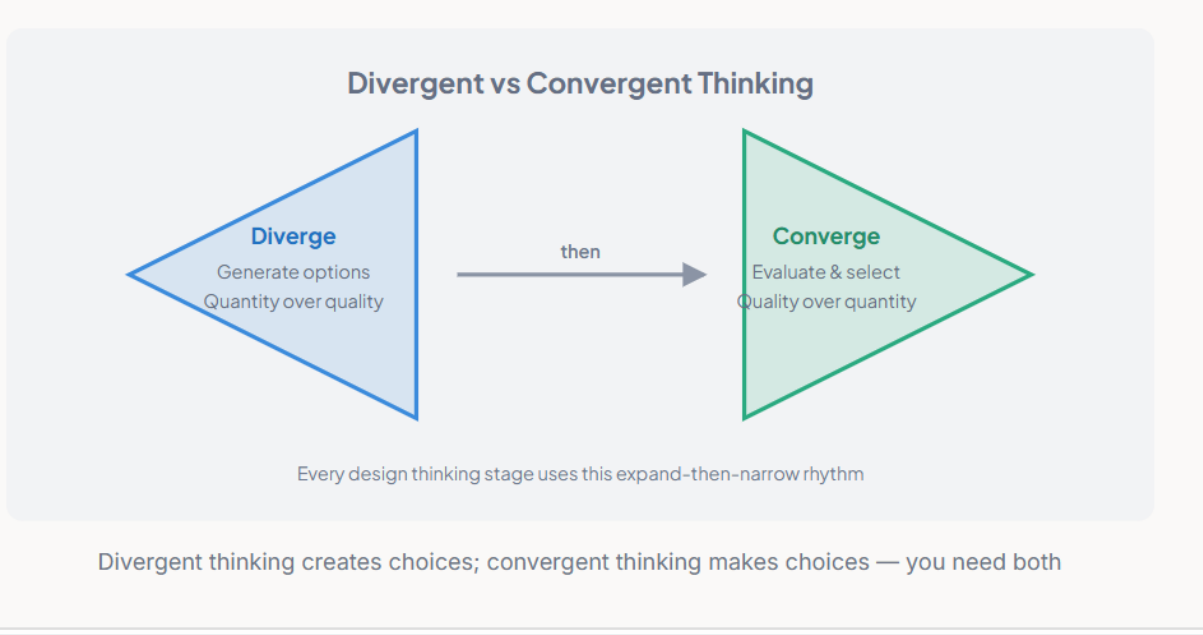
Divergent and convergent in Design Thinking

Every design thinking project swings between two fundamentally different cognitive modes. One asks you to generate as many possibilities as possible. The other asks you to narrow down to the best option. Getting this rhythm wrong is one of the most common reasons teams stall partway through a project, yet most guides treat these modes as background knowledge rather than something you can deliberately practice and improve.

By Keith Li — Design thinking educator



Divergent thinking creates choices; convergent thinking makes choices — you need both

What Divergent Thinking Actually Means

Divergent thinking is the act of expanding the solution space. You are not looking for the right answer; you are looking for many answers. The goal is volume and variety. A good divergent session produces ideas that surprise even the people who came up with them.

In practice, divergent thinking shows up in the Empathize stage when you explore multiple user segments before deciding who to focus on, and again in the Ideate stage when you brainstorm solutions. But it also appears in smaller moments: when you draft three different problem statements instead of one, or when you sketch five layout variations before committing.

The psychological requirement for divergent thinking is suspension of judgment. The moment someone in the room says "that will never work," the group shifts into evaluation mode and the divergent phase ends prematurely. This is why techniques like structured brainstorming and Crazy 8s impose rules that physically prevent premature critique.

What Convergent Thinking Actually Means

Convergent thinking is the act of narrowing the solution space. You take the broad set of options generated during divergence and apply criteria, constraints, and judgment to select the most promising ones. Where divergence asks "what could we do?", convergence asks "what should we do?"

Convergent thinking requires explicit criteria. Without them, the loudest voice in the room wins. Tools like dot voting and impact/effort matrices exist specifically to make convergence more democratic and evidence-based.

A common mistake is treating convergence as a single event. In reality, you converge multiple times: first from dozens of ideas to a shortlist, then from the shortlist to a concept, then from concept variations to a prototype specification. Each round uses tighter criteria than the last.

How the Two Modes Map to Design Thinking Stages

The relationship between divergent and convergent thinking is not a simple "first one, then the other." Each stage of design thinking contains its own internal cycle of expansion and contraction. Understanding this pattern helps you recognize where you are and what kind of thinking the moment demands.

Stage	Divergent Phase	Convergent Phase
Empathize	Interview many users, observe multiple contexts, gather broad qualitative data	Synthesize into key themes, build focused empathy maps and personas
Define	Write multiple HMW questions, explore different problem framings	Select the single most actionable problem statement
Ideate	Generate as many solution ideas as possible without filtering	Evaluate, cluster, and select ideas worth prototyping
Prototype	Build multiple low-fidelity versions exploring different directions	Choose the prototype that best tests the riski assumption

Test	Collect broad feedback from diverse users, note unexpected reactions	Decide what to iterate, pivot, or ship based o patterns

A **Morphological Chart** is a **design-thinking and engineering design tool** used to generate many possible solutions by **combining different alternatives for different functions or features** of a product/system.

Example: Smart Teaching-Learning System:

Suppose we want to design a **Smart Teaching-Learning System**.

Function	Alternative 1	Alternative 2	Alternative 3
User Login	Email	Mobile OTP	College ID
Learning Material	PDF	Video	Interactive Content
Communication	Email	Chat	Discussion Forum
Assessment	MCQ	Assignment	Online Quiz
Feedback	Rating	Comments	Survey

We can select one option from each row:

Concept 1:
Email Login + PDF + Email + MCQ + Rating

Concept 2:
Mobile OTP + Video + Chat + Online Quiz + Survey

Concept 3:
College ID + Interactive Content + Discussion Forum + Assignment + Comments

Thus, the morphological chart helps us **systematically explore different combinations instead of choosing only one solution immediately.**

Morphological chart Vs Mind map

Morphological Chart	Mind Map
Used to generate design alternatives	Used to organize and explore ideas
Usually represented as a table	Usually represented as branches
Combines alternatives to create concepts	Shows relationships between ideas
Common in product/system design	Common in brainstorming and ideation

A **mind map** is a **visual diagram** used to organize ideas, information, and concepts around a **central topic**. It starts with the main idea in the center, and related ideas branch out into subtopics, making it easier to brainstorm, analyze, and understand relationships.

In **Design Thinking**, mind mapping is an **ideation technique** used to generate, organize, and connect ideas before selecting the best solution.

Mindmap :

Starts with a **central idea**.
Uses **branches** for main topics.
Includes **sub-branches** for detailed ideas.
Shows relationships between concepts.
Encourages creative thinking and brainstorming.
Easy to understand and modify.

Purpose of a Mind Map

- Organize ideas visually.
- Generate creative solutions.
- Identify relationships between concepts.
- Improve problem-solving.
- Support planning and decision-making.

Applications

Mind maps are used in:

- Design Thinking
- Software Engineering
- Project Planning
- Education
- Business Strategy
- Research and Problem Solving